

Basic Comparison Test or Direct Comparison Test

Basic Comparison Test for Infinite Series $\sum_{n=1}^{\infty} a_n$ where $a_n > 0$

Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are positive-term series.

Then:

1) If $\sum_{n=1}^{\infty} b_n$ converges and $a_n \leq b_n$ for all n , then $\sum_{n=1}^{\infty} a_n$ converges.

2) If $\sum_{n=1}^{\infty} b_n$ diverges and $b_n \leq a_n$ for all n , $\sum_{n=1}^{\infty} a_n$ diverges.

$$1) \sum_{n=1}^{\infty} \frac{1}{2n-1}$$

Let $a_n = \frac{1}{2n-1}$ and $b_n = \frac{1}{2n}$ Note: $2n > 2n-1 > 0$; Hence, $0 < \frac{1}{2n} < \frac{1}{2n-1}$.

a) Explain why the series $\sum_{n=1}^{\infty} \frac{1}{2n} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n}$ diverges?

Note: $\sum_{n=1}^{\infty} \frac{1}{2n} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n}$ is a harmonic series or p-series with $p=1$

b) $\sum_{n=1}^{\infty} \frac{1}{2n} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n} = \underline{\hspace{2cm}}$ (finite value or ∞ ?)

c) Compare $\sum_{n=1}^{\infty} \frac{1}{2n} \underline{\hspace{0.5cm}} \sum_{n=1}^{\infty} \frac{1}{2n-1}$ ($<$, $>$, $=$?)

d) Use the Basic Comparison Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{2n-1}$ diverges?

$$2) \sum_{n=2}^{\infty} \frac{1}{\sqrt{n}-1}$$

Let $a_n = \frac{1}{\sqrt{n}-1}$ and $b_n = \frac{1}{\sqrt{n}}$ Note: $\sqrt{n} > \sqrt{n}-1 \geq 0$ for $n \geq 2$ Hence, $0 \leq \frac{1}{\sqrt{n}} < \frac{1}{\sqrt{n}-1}$ for $n \geq 2$

a) Explain why the series $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ diverges. Note: $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ is a p-series with $p=1/2$.

b) $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}} = \underline{\hspace{2cm}}$ (finite value or ∞ ?)

c) Compare $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}}$? $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}-1}$ ($<$, $>$, $=$?)

d) Use the Basic Comparison Test to explain why the series $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}-1}$ diverges.

$$3) \sum_{n=1}^{\infty} \frac{\ln n}{n+1}$$

$$\text{Note: } 2n > n+1 \Rightarrow \frac{\ln n}{2n} < \frac{\ln n}{n+1} \Rightarrow \sum_{n=1}^{\infty} \frac{\ln n}{2n} < \sum_{n=1}^{\infty} \frac{\ln n}{n+1}$$

$$\text{a) Find } \int_1^{\infty} \frac{\ln x}{2x} dx = \frac{1}{2} \cdot \int_1^{\infty} \frac{\ln x}{x} dx = \underline{\hspace{2cm}} \quad \text{Hint: Let } u = \ln x$$

$$\text{b) Use the Integral Test to explain why the } \sum_{n=1}^{\infty} \frac{\ln n}{2n} \text{ diverges.}$$

$$\text{c) Use the Basic Comparison Test to explain why the } \sum_{n=1}^{\infty} \frac{\ln n}{n+1} \text{ diverges.}$$

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$$4) \sum_{n=0}^{\infty} \frac{1}{n!}$$

$$\text{Hint: Let } a_n = \frac{1}{n!} \text{ and } b_n = \frac{1}{n^2}$$

Note: For $n > 3$, $n! > n^2$; Hence, $0 < \frac{1}{n!} < \frac{1}{n^2}$ and $\sum_{n=4}^{\infty} \frac{1}{n!} < \sum_{n=4}^{\infty} \frac{1}{n^2}$ for $n > 3$

$\sum_{n=4}^{\infty} \frac{1}{n^2}$ is a p-series with $p = 2$

$$\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{3} + \sum_{n=4}^{\infty} \frac{1}{n!} < 1 + 1 + \frac{1}{2} + \frac{1}{3} + \sum_{n=4}^{\infty} \frac{1}{n^2}$$

a) Explain why the series $\sum_{n=4}^{\infty} \frac{1}{n^2}$ converges.

b) $\sum_{n=4}^{\infty} \frac{1}{n^2} = \underline{\hspace{2cm}}$ (finite value or ∞ ?)

c) $\sum_{n=0}^{\infty} \frac{1}{n!} = \underline{\hspace{2cm}}$ (finite value or ∞ ?)

d) Use the Basic Comparison Test to explain why the series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges.

$$5) \sum_{n=0}^{\infty} \frac{1}{e^{n^2}}$$

$$\text{Hint: Let } a_n = \frac{1}{e^{n^2}} \quad \text{and} \quad b_n = \frac{1}{e^n}$$

$$\text{Note: } e^{n^2} > e^n; \text{ Hence, } 0 < \frac{1}{e^{n^2}} < \frac{1}{e^n} \quad \text{and} \quad \sum_{n=0}^{\infty} \frac{1}{e^{n^2}} \leq \sum_{n=0}^{\infty} \frac{1}{e^n}$$

$$\text{Note: } \sum_{n=0}^{\infty} \frac{1}{e^n} = \sum_{n=0}^{\infty} \left(\frac{1}{e}\right)^n \text{ is a Geometric Series with } r = \frac{1}{e}$$

$$\text{a) Explain why the series } \sum_{n=0}^{\infty} \left(\frac{1}{e}\right)^n \text{ converges.}$$

$$\text{b) } \sum_{n=0}^{\infty} \frac{1}{e^{n^2}} = \text{_____ (finite value or } \infty?)$$

$$\text{c) } \sum_{n=0}^{\infty} \frac{1}{e^n} = \text{_____ (finite value or } \infty?)$$

$$\text{d) Use the Basic Comparison Test to explain why the series } \sum_{n=0}^{\infty} \frac{1}{e^{n^2}} \text{ converges.}$$

$$6) \sum_{n=1}^{\infty} \frac{n}{n^2 + 1}$$

Hint: Let $a_n = \frac{n}{n^2 + 1}$ and $b_n = \frac{n}{2n^2}$

Note: $2n^2 \geq n^2 + 1 \Rightarrow \frac{n}{n^2 + 1} \geq \frac{n}{2n^2} = \frac{1}{2n} \Rightarrow \sum_{n=1}^{\infty} \frac{n}{n^2 + 1} \geq \sum_{n=1}^{\infty} \frac{1}{2n}$

a) Find $\int_1^{\infty} \frac{1}{2x} dx = \frac{1}{2} \cdot \int_1^{\infty} \frac{1}{x} dx = \underline{\hspace{2cm}}$

b) Use the Integral Test to explain why the $\sum_{n=1}^{\infty} \frac{1}{2n}$ diverges.

c) Use the Basic Comparison Test to explain why the $\sum_{n=1}^{\infty} \frac{n}{n^2 + 1}$ diverges.

$$7) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2 + 1}}$$

Hint: Let $a_n = \frac{1}{\sqrt{n^2 + 1}}$ and $b_n = \frac{1}{\sqrt{2n^2}}$

Note: $2n^2 \geq n^2 + 1 \Rightarrow \frac{1}{\sqrt{n^2 + 1}} \geq \frac{1}{\sqrt{2n^2}} = \frac{1}{\sqrt{2} \cdot n} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2 + 1}} \geq \sum_{n=1}^{\infty} \frac{1}{\sqrt{2} \cdot n}$

a) Find $\int_1^{\infty} \frac{1}{\sqrt{2} \cdot x} dx = \frac{1}{\sqrt{2}} \cdot \int_1^{\infty} \frac{1}{x} dx = \underline{\hspace{2cm}}$

b) Use the Integral Test to explain why the $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2} \cdot n}$ diverges.

c) Use the Basic Comparison Test to explain why the $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2 + 1}}$ diverges.

8) $\sum_{n=1}^{\infty} \frac{1}{5^n + 1}$

Hint: Let $a_n = \frac{1}{5^n + 1}$ and $b_n = \frac{1}{5^n}$

Note: $\frac{1}{5^n + 1} \leq \frac{1}{5^n} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{5^n + 1} \leq \sum_{n=1}^{\infty} \frac{1}{5^n}$

b) Use the Geometric Series Test to explain why the $\sum_{n=1}^{\infty} \frac{1}{5^n}$ converges.

c) Use the Basic Comparison Test to explain why the $\sum_{n=1}^{\infty} \frac{1}{5^n + 1}$ converges.

